

DBMS Module 1 – Easy & Detailed Notes

MODULE 1 – DATABASE MANAGEMENT SYSTEM (DBMS)

1. Introduction to Database

What is Data?

Data means raw facts and figures.

Data alone may not give proper meaning.

Examples:

- 10
- ABC
- 95
- Rahul
- 9876543210

These are only raw values.

What is Information?

When data is organized properly and gives meaning, it becomes information.

Example:

Name	Phone Number	City
Rahul	9876543210	Bangalore

Now the data becomes meaningful.

What is a Record?

A collection of related data is called a Record.

Example:

USN	Name	Marks
101	Ravi	85

The complete row is called a Record.

Important Table Terms

Term	Meaning
Row	Record / Tuple
Column	Field / Attribute
Table	Relation
Database	Collection of tables

What is a Database?

A database is an organized collection of related data.

It stores data in a structured manner.

Examples of Databases:

- College database
 - Bank database
 - Hospital database
 - Railway reservation database
 - Library database
-

Properties of Database**1. Represents Real World**

Database represents real-world objects.

Example:

Student database represents real students.

2. Organized Collection

Data is stored logically and meaningfully.

Random data is not called a database.

3. Designed for Specific Purpose

Every database is created for some purpose.

Example:

Bank database is used for account management.

4. Database Can Be Small or Large

Small database:

- Personal contact list

Large database:

- Facebook
 - Railway reservation system
-

2. DBMS (Database Management System)

What is DBMS?

DBMS is software used to create and manage databases.

It helps users to:

- store data
 - retrieve data
 - update data
 - delete data
 - manage security
-

Examples of DBMS

- MySQL
- Oracle
- SQL Server

- PostgreSQL
 - MongoDB
-

Functions of DBMS

1. Defining Database

DBMS defines:

- tables
- attributes
- data types
- constraints

Example:

Creating STUDENT table.

2. Constructing Database

DBMS stores data into storage devices.

Example:

Saving student details into hard disk.

3. Manipulating Database

Manipulation means:

- insert
- update
- delete
- search

Example:

Updating student marks.

4. Sharing Database

Many users can access same database together.

Example:

Many railway booking agents using same reservation system.

Database System

Database + DBMS software together form a Database System.

3. File System vs Database System

File Processing System

Old method of storing data.

Each department stores separate files.

Problems:

- duplicate data
- difficult searching
- poor security
- data inconsistency
- difficult backup

Example:

Student data stored separately in:

- library
- exam section
- office

Same data repeated many times.

Database Approach

Data is stored centrally.

Advantages:

- less redundancy
- better consistency
- easier access
- improved security

- easier maintenance
-

4. Characteristics of DBMS

1. Self-Describing Nature

DBMS stores:

- actual data
- database structure
- constraints

This information is called Metadata.

Metadata

Metadata means data about data.

Example:

Information about:

- table names
 - column names
 - data types
-

2. Program-Data Independence

Programs are independent from database structure.

If database structure changes, programs usually need not change.

Example:

Adding a new column in table without changing all applications.

3. Data Abstraction

Users do not need to know how data is stored internally.

They only see required information.

Example:

ATM user sees account balance but not storage details.

4. Multiple Views of Data

Different users can see different views.

Example:

User	View
Student	Marks
Teacher	Attendance + Marks
Admin	Complete Details

5. Multiuser Transaction Processing

Many users can access database simultaneously.

DBMS controls access correctly.

Example:

Two users should not reserve same seat.

Transaction Properties

1. Atomicity

Either complete transaction happens or nothing happens.

Example:

Money transfer:

- debit and credit both should happen
 - if one fails, entire transaction cancels
-

2. Isolation

Transactions work independently.

One transaction should not disturb another.

5. Users of DBMS

1. Database Administrator (DBA)

Responsible for:

- security
- backup
- recovery
- permissions
- performance tuning

Example:

Giving login access to users.

2. Database Designers

Design database structure.

Responsibilities:

- identify data
 - create tables
 - define relationships
-

3. End Users

People who use database.

Types of End Users

Casual Users

Use database occasionally.

Example:

Managers.

Naive Users

Use predefined applications.

Examples:

- ATM users
 - Railway booking clerks
 - Bank tellers
-

Sophisticated Users

Expert users.

Examples:

- engineers
 - analysts
 - scientists
-

Standalone Users

Maintain personal databases.

Example:

Personal finance software users.

4. System Analysts

Analyze user requirements.

5. Application Programmers

Develop database applications.

6. Advantages of DBMS

1. Controls Data Redundancy

Duplicate data is reduced.

Example:

Student address stored only once.

2. Better Security

Unauthorized users can be blocked.

Example:

Only admin can modify salary details.

3. Backup and Recovery

Data can be recovered after failure.

Example:

Recovering data after system crash.

4. Multiple User Interfaces

Different interfaces for different users.

Examples:

- forms
 - menus
 - graphical interfaces
-

5. Data Integrity

Maintains correct data.

Example:

Age cannot be negative.

6. Efficient Data Access

Searching becomes faster.

7. Better Data Sharing

Many users can share same data.

8. Represents Relationships Easily

Relationships between tables can be maintained.

Example:

Student linked with department.

7. Disadvantages of DBMS

- High cost
 - Complex system
 - Requires training
 - Hardware requirements are high
 - Extra overhead for security and recovery
-

8. Data Models

What is Data Model?

A data model describes:

- structure of data
 - relationships
 - constraints
-

Types of Data Models

1. High-Level / Conceptual Data Model

Easy for users to understand.

Uses:

- entities
- attributes
- relationships

Example:

ER Model.

2. Physical Data Model

Describes physical storage.

Focuses on:

- files
- disks
- indexes

Used by technical experts.

3. Representational / Relational Model

Data represented using tables.

Most widely used model.

Example:

Student_ID	Name
1	Ravi

9. Schema and Instance

Schema

Schema means database structure/design.

Example:

STUDENT(RollNo, Name, Age)

Schema changes rarely.

Instance (Snapshot)

Actual data stored at a particular time.

Example:

RollNo	Name
1	Ravi
2	Priya

Instance changes frequently.

Difference Between Schema and Instance

Schema	Instance
Structure	Actual data
Rarely changes	Changes frequently

Schema Evolution

Changing database schema.

Example:

Adding new attribute: Date_of_Birth

10. Three Schema Architecture

Purpose: Separate users from physical storage.

1. Internal Level

Lowest level.

Describes:

- physical storage
 - files
 - indexes
 - storage details
-

2. Conceptual Level

Middle level.

Describes:

- complete database structure
 - entities
 - relationships
 - constraints
-

3. External/View Level

Highest level.

Each user sees only required data.

Example:

Teacher sees marks. Student sees own result.

Data Independence

Logical Data Independence

Changing conceptual schema without changing external views.

Example:

Adding new field to table.

Physical Data Independence

Changing storage details without affecting users.

Example:

Changing indexing method.

11. Database Languages

1. DDL (Data Definition Language)

Used to define database structure.

Commands:

- CREATE
- ALTER
- DROP

Example:

```
CREATE TABLE STUDENT;
```

2. DML (Data Manipulation Language)

Used to manipulate data.

Commands:

- INSERT
- UPDATE
- DELETE
- SELECT

Example:

```
INSERT INTO STUDENT VALUES(1,'Ravi');
```

3. SDL (Storage Definition Language)

Defines storage structure.

4. VDL (View Definition Language)

Defines user views.

SQL

SQL stands for Structured Query Language.

Used in most DBMS.

Types of DML

High-Level DML

Easy to use. Works on multiple records.

Example:

```
SELECT * FROM STUDENT;
```

Low-Level DML

Works record by record. Usually embedded in programming languages.

Difference Between DDL and DML

DDL	DML
Defines structure	Manipulates data
CREATE	INSERT
ALTER	UPDATE
DROP	DELETE

12. DBMS Interfaces

1. Menu-Based Interface

User selects options from menus.

Example:

ATM machine.

2. Forms-Based Interface

Users fill forms.

Example:

Student registration form.

3. Graphical User Interface (GUI)

Uses graphics and icons.

Example:

phpMyAdmin.

4. Natural Language Interface

Accepts normal English commands.

Example:

"Show all students with marks above 90"

5. Speech Interface

Uses voice input/output.

6. DBA Interface

Used by database administrators.

13. DBMS Components

1. Query Processor

Processes SQL queries.

2. Query Optimizer

Chooses best execution method.

Improves speed.

3. Storage Manager

Controls stored data.

4. Buffer Manager

Manages memory buffers.

5. Backup and Recovery Manager

Recovers data after failure.

14. Database Utilities

1. Loading Utility

Loads external data into database.

2. Backup Utility

Creates backup copies.

3. Reorganization Utility

Improves database performance.

4. Performance Monitoring Utility

Checks database performance.

15. ER Model (Entity Relationship Model)

ER Model is used for database design.

Main concepts:

- Entity
 - Attribute
 - Relationship
-

16. Entity

Entity is a real-world object.

Examples:

- Student
 - Employee
 - Car
 - Course
-

17. Attributes

Attributes describe entities.

Example:

EMPLOYEE:

- Name
 - Age
 - Salary
-

Types of Attributes

1. Simple Attribute

Cannot be divided.

Example:

Age

2. Composite Attribute

Can be divided into smaller parts.

Example:

Address → Street, City, State

3. Single-Valued Attribute

Only one value.

Example:

SSN

4. Multi-Valued Attribute

Can have many values.

Example:

Phone Numbers

5. Stored Attribute

Stored directly in database.

Example:

Birth Date

6. Derived Attribute

Calculated from another attribute.

Example:

Age from Birth Date.

7. NULL Value

Value missing or unknown.

Example:

Apartment number for independent house.

8. Complex Attribute

Combination of composite and multivalued attributes.

18. Entity Type and Entity Set

Entity Type

Collection of similar entities.

Example:

EMPLOYEE

Entity Set

Collection of all entity instances.

Example:

All employees in company.

19. Key Attribute

Attribute used to uniquely identify entities.

Example:

Employee_ID USN SSN

Properties:

- unique
 - no duplicates
-

Weak Entity

Weak entity does not have its own key.

Depends on another entity.

Example:

DEPENDENT depends on EMPLOYEE.

Difference Between Strong and Weak Entity

Strong Entity	Weak Entity
Has own key	No own key
Independent	Depends on another entity

20. Relationship

Relationship connects entities.

Example:

EMPLOYEE works for DEPARTMENT.

Relationship Types

Binary Relationship

Between two entities.

Example:

STUDENT — COURSE

Ternary Relationship

Between three entities.

Example:

SUPPLIER — PART — PROJECT

Recursive Relationship

Entity related to itself.

Example:

EMPLOYEE supervises EMPLOYEE.

21. Cardinality Constraints

Shows maximum number of relationships.

Types of Cardinality

1:1 (One to One)

One entity connected to only one entity.

Example:

One department has one manager.

1:N (One to Many)

One entity connected to many entities.

Example:

One department has many employees.

M:N (Many to Many)

Many entities connected to many entities.

Example:

Students enrolled in many courses.

22. Participation Constraints

Total Participation

Every entity must participate.

Example:

Every employee must belong to a department.

Partial Participation

Participation is optional.

Example:

Not every employee is a manager.

23. Attributes of Relationships

Relationships can also have attributes.

Example:

WORKS_ON relationship may contain:

- Hours
 - Start_Date
-

24. Naming Rules in ER Model

- Use meaningful names
- Use singular names
- Entity names in uppercase
- Attribute names start with capital letters

Example:

EMPLOYEE Department_Name

25. UML Class Diagram

In UML:

- Entity → Class
- Entity instance → Object

Class contains:

- class name
 - attributes
 - operations
-

26. Important Quick Revision

Database

Collection of related data.

DBMS

Software used to manage database.

Metadata

Data about data.

Schema

Structure of database.

Instance

Current data.

Entity

Real-world object.

Attribute

Property of entity.

Relationship

Connection between entities.

SQL

Database query language.

27. Very Important Exam Topics

- Advantages and disadvantages of DBMS
- File system vs DBMS
- Data models
- Three-schema architecture

- Data independence
 - DDL and DML
 - ER model
 - Types of attributes
 - Cardinality constraints
 - Weak entity
 - Schema and instance
-

28. Easy Memory Tricks

Functions of DBMS

D-C-M-S

- Define
 - Construct
 - Manipulate
 - Share
-

Three Schema Levels

E-C-I

- External
 - Conceptual
 - Internal
-

Languages

DDL → Define DML → Manipulate VDL → View SDL → Storage

End of Easy & Detailed DBMS Module 1 Notes